State Machines and the LCD Display

Module	ET4047
Date	28/10/2026
Lab No.	Lab 6
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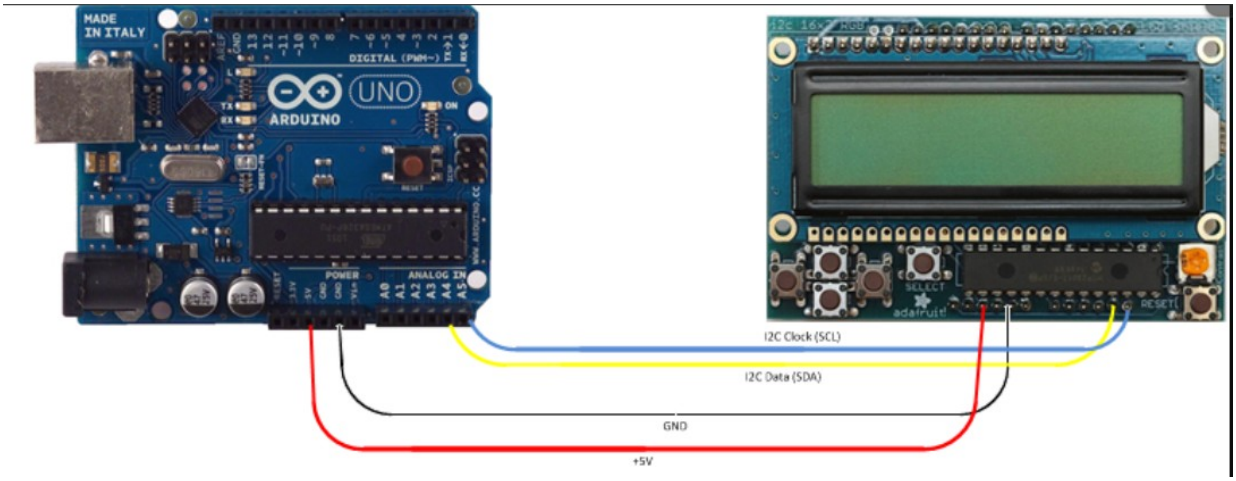


Figure 1: Adafruit RGB LCD shield wiring: **5V**, **GND**, **SDA → A4** and **SCL → A5**

1 Task 1: A traffic light on the LCD

Objective: run a three-state traffic light (RED → GREEN → YELLOW) and change the state machine.

Wiring

I2C LCD connections (used in every task)

From		To	Purpose
LCD **GND**	→	Nano **GND**	Ground
LCD **VCC**	→	Nano **5V**	Power
LCD **SDA**	→	Nano **A4**	I2C data
LCD **SCL**	→	Nano **A5**	I2C clock

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LCD wiring (every task)

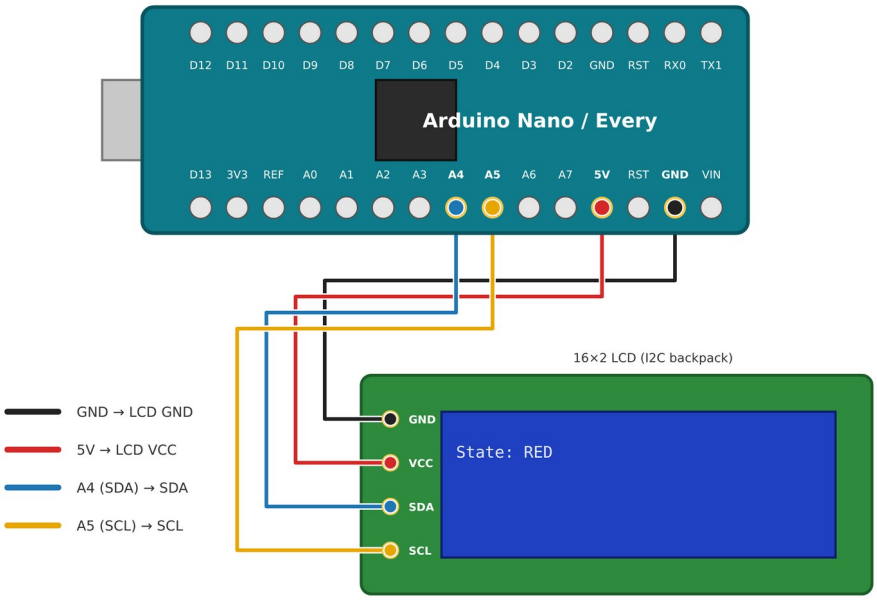


Figure 2: I2C backpack LCD wiring. For the RGB LCD shield, see Figure 1.

Task 1 state diagram: the traffic light

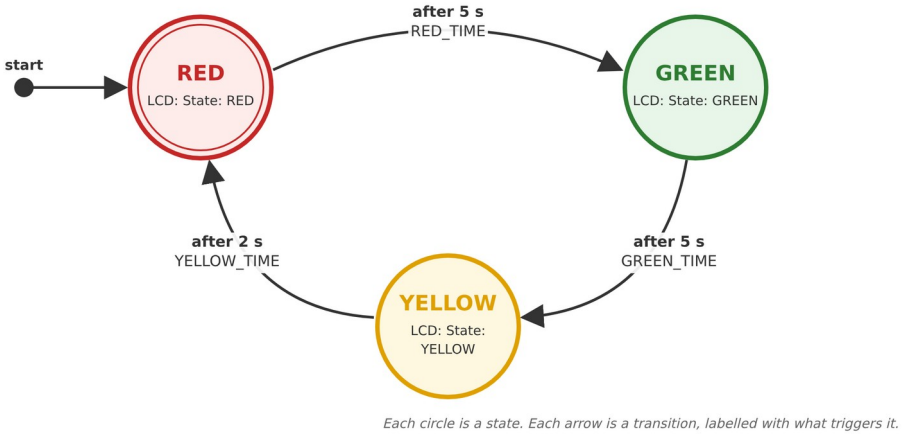
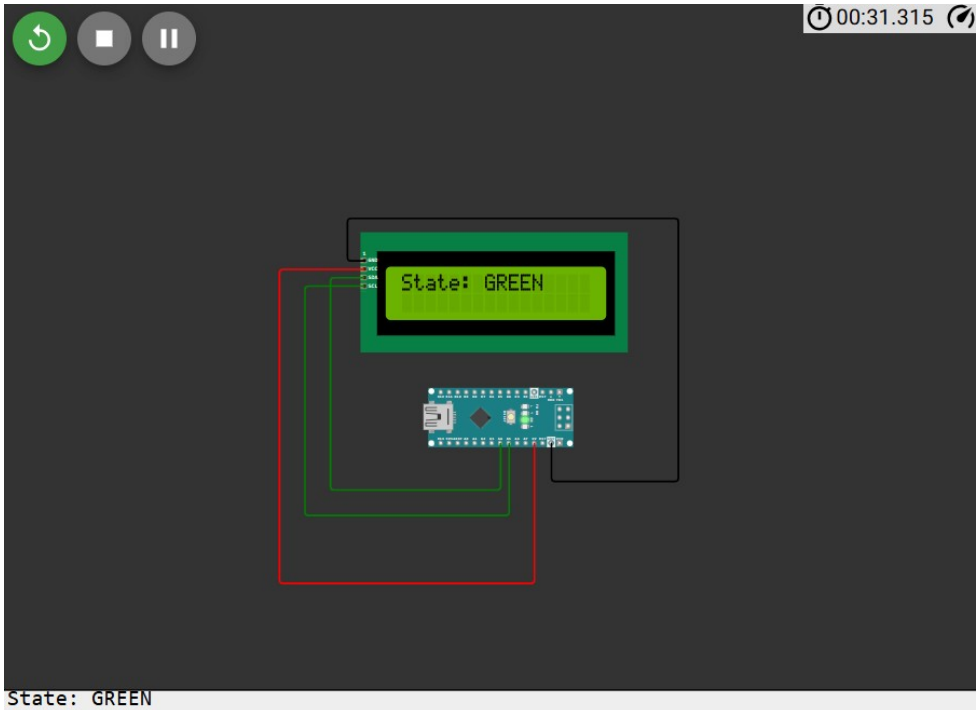


Figure 3: Task 1 state diagram

> **Code**
>
> Upload `Task1_Traffic_Light`. Use the folder for your display: `I2C_Backpack_LCD` (separate screen with four wires) or `Adafruit_RGB_LCD_Shield` (the shield with five buttons).

Expected result: the LCD shows `State: RED` for 5 s, `State: GREEN` for 5 s, `State: YELLOW` for 2 s, and repeats.

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```
State: GREEN
State: YELLOW
State: RED
State: GREEN
State: YELLOW
State: RED
State: GREEN
```

Figure 4: Task 1 running in the Wokwi simulator, with the Serial Monitor underneath

Task 1 questions

1a Run it

Upload the code and check the states change in the right order. **(record)** Take a photo of the LCD.

1b The transition table

Fill in the table from the code.

Current state	Time in state	Next state	LCD shows
RED			
GREEN			
YELLOW			

How long does one full cycle take?

Your answer:

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1c Change the timing

Make YELLOW last 3 seconds instead of 2. Which line did you change? How long is one full cycle now?

Your answer:

1d Add a state

Add a fourth state, `RED_AMBER`, which shows `State: RED+AMBER` for 2 s, so the cycle becomes RED → RED_AMBER → GREEN → YELLOW → RED. Which three places in the code did you change? **(record)** Draw your new state diagram.

Your answer:

2 Task 2: Traffic light with real LEDs

Objective: drive a red, a yellow and a green LED from the state machine, with exactly one LED on at a time.

Wiring

LED connections (keep the LCD wiring from Task 1)

From		To	Purpose
Nano **D5**	→	220 Ω → **red** LED long leg	Red light
Nano **D6**	→	220 Ω → **yellow** LED long leg	Yellow light
Nano **D7**	→	220 Ω → **green** LED long leg	Green light
Each LED short leg	→	GND rail → Nano **GND**	Common ground

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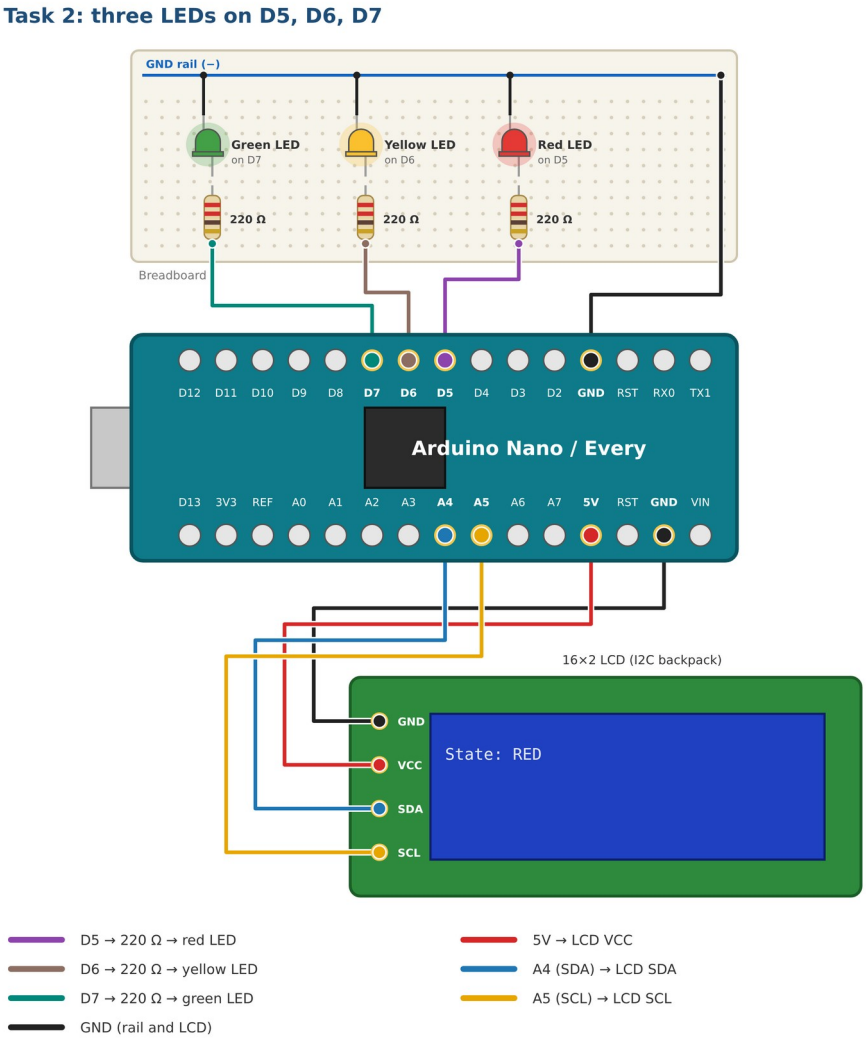


Figure 5: Task 2 wiring

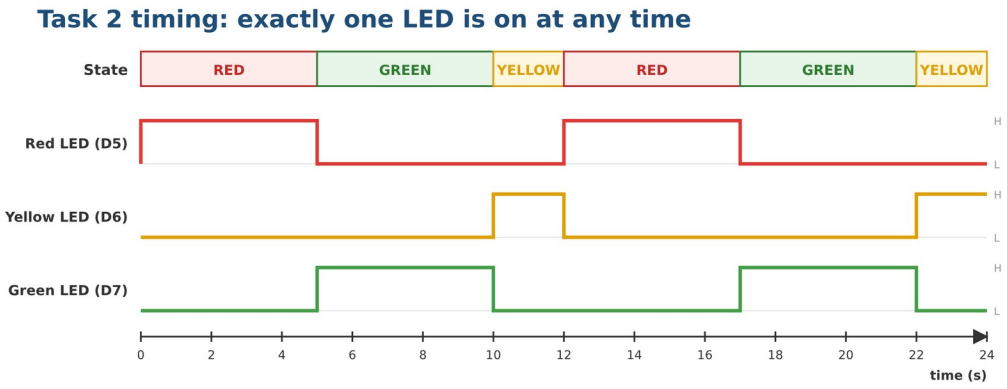


Figure 6: Task 2 LED outputs over two cycles

Code

Upload `Task2_Traffic_Light_LEDs`. Use the folder for your display: `I2C_Backpack_LCD` (separate screen with four wires) or `Adafruit_RGB_LCD_Shield` (the shield with five buttons).

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Expected result: one LED at a time, in step with the LCD.

Task 2 questions

2a Run it

(record) Paste a few lines from the Serial Monitor. They should look like this:

Serial Monitor

```
State: RED
State: GREEN
State: YELLOW
```

2b Forget to switch an LED off

In every case, delete the two lines that switch the other LEDs `LOW`. Run it for two cycles. What do the LEDs show, and why? Compare with Figure 6. Then put the lines back.

Your answer:

2c The resistor

A red LED drops about 2 V and the pin gives 5 V. Roughly how much current flows through the 220 Ω resistor?

Your answer:

3 Task 3: A pedestrian crossing

Objective: make a transition happen because of an **input**. The light now stays GREEN until a pedestrian presses the button, then goes YELLOW → RED as before.

Wiring

Add the button (keep the LEDs and LCD from Task 2)

From		To	Purpose
Nano **D2**	→	Push button, one leg	Pedestrian button
Push button, other leg	→	GND rail	Pressing pulls D2 to 0 V

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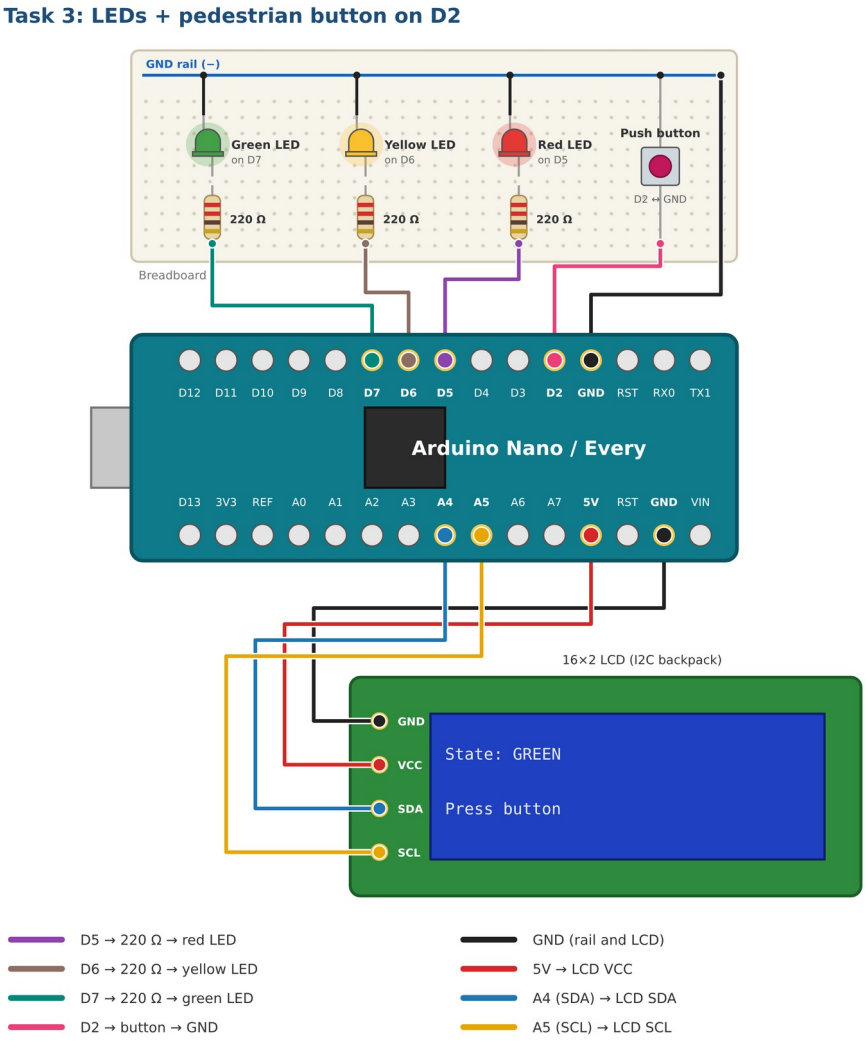


Figure 7: Task 3 wiring

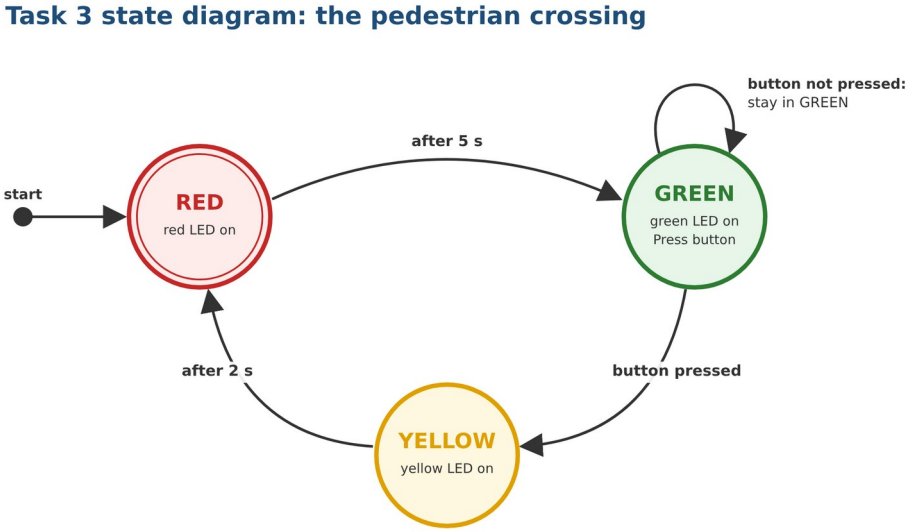


Figure 8: Task 3 state diagram

> **Code**
>
> Upload `Task3_Pedestrian_Crossing`. Use the folder for your display: `I2C_Backpack_LCD` (separate screen with four wires) or `Adafruit_RGB_LCD_Shield` (the shield with five buttons).

Expected result: RED for 5 s, then GREEN with `Press button` on the LCD until you press the button, then YELLOW for 2 s and back to RED.

Task 3 questions

3a Run it

(record) Paste the Serial Monitor output for one cycle. It should look like this:

Serial Monitor

```
State: RED
State: GREEN
Button pressed
State: YELLOW
```

3b Press it at different times

Press the button during RED, during GREEN and during YELLOW. What happens each time, and why?

Your answer:

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3c Waiting in a state

In `case GREEN:` the `while` loop waits for the button. Could the board flash an LED or check a second button while it waits there? Why not?

Your answer:

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